Ellen Ma

ellenma@g.harvard.edu | ellenma.vercel.app |

EDUCATION

Harvard University *M.S. Data Science* **Sept. 2024 – Jan. 2026**

- Thesis: "Simulator-Augmented LLM Agents for Multi-Step Planning"
- Advisor: Kianté Brantley

University of Chicago *B.S. Mathematics, B.A. Economics* **Sept. 2020 – Jan. 2024**

- Dean's List, Katen scholar

WORK EXPERIENCE

Metis **Feb. 2026 – Present**

ML Researcher

- Trained LLM agents via GRPO on a multi-domain consumer recommendation benchmark requiring live web search and browsing tools for grounded citations
- Diagnosed cross-domain performance gaps and iterated on reward integration and SFT data curation strategies
- Built and debugged distributed RL training stack: VeRL with SGLang rollouts, FSDP2+LoRA, including patches for PEFT compatibility
- Optimized LLM judge prompts via GEPA for evaluating generated merchant profiles, aligning automated scores with human labels

Kempner Institute at Harvard University **Jun. 2025 – Jan. 2026**

ML Research Intern

- Designed meta-MDP framework enabling LLMs to choose between direct actions and simulator tool calls for planning without mutating environment state in Frozenlake, Sokoban ASCII grid worlds
- Built SFT data pipeline with BFS-solved chain-of-thought reasoning traces, deadlock injection for negative examples, and difficulty-parameterized curriculum
- Configured multi-node Ray on FASRC H100 clusters for OpenRLHF and VeRL; created custom multi-turn Gym wrappers for asynchronous RL training
- Stabilized tool-use training via GRPO/PPO ablations on KL penalties, reward shaping, and intrinsic reward capping to mitigate reward hacking
- Built evaluation harness executing LLM actions in live Gym environments over multi-turn rollouts, scoring rewards (agents learned strategic simulator use on FrozenLake; identified collapse failures on Sokoban)

Garcia Lab at UChicago Medicine **Jan. 2021 – Dec. 2023**

Research Assistant

- Built signal processing and ML pipeline (RandomForest, PCA, UMAP, FFT) to analyze ventilatory reflexes and detect changes in preBötzinger complex activity (brainstem respiratory control center)
- Findings revealed physiological adaptations that may explain variance in overdose susceptibility

1104Health **May 2022 – Sept. 2022**

Data Science Intern

- Applied NLP techniques including Named Entity Recognition, relation extraction, and negation detection on medical terminology and healthcare data to distinguish inclusion/exclusion criteria in clinical trials
- Presented literature reviews and visualized data to highlight inefficiencies in clinical trial matching and demand for product to partners and investors (e.g. Johns Hopkins Medicine)

SKILLS

- **Programming:** Python, R, PyTorch, Git, GCP, Docker, Slurm, sklearn, NumPy, pandas
- **ML/AI:** reinforcement learning, supervised fine-tuning, distributed training, Ray, Slurm, OpenRLHF, VeRL
- **Coursework:** differential equations, probability theory, Markov chains & Brownian motion, machine learning